

Oscillation of the optimal value in the all-heads coin game

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Abstract

In the all-heads coin game a player tosses a pool of n coins, each landing heads with probability p , must set aside at least one head each round, and loses on an all-tails round; $w_{n,p}$ denotes the optimal winning probability. We show — modulo one explicitly identified gap — that for every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone, with infinitely many strict local maxima and minima. The proof reduces the claim to two one-sided statements: the increasing side is settled by an elementary Bellman estimate; the decreasing side is reduced, by an exact transfer to the greedy strategy and a renormalisation argument across geometric scales, to a single non-degeneracy condition (Gap D). The log-periodic asymptotics of the greedy strategy that drive the argument are established in the companion paper [2]; the exact and machine-checked analysis of the game is [1]. This is a working draft: Gap D is the one step that remains open.

*Working draft. The one step still to be completed is flagged in the text by a boxed **OPEN GAP** marker (Gap D); Section 8 consolidates its status.*

1 Introduction

1.1 The game

Fix $p \in (0, 1)$ and write $q := 1 - p$. A player holds n coins, each of which lands heads independently with probability p . In each round all remaining coins are tossed; the player must set aside at least one coin showing heads, and loses if no coin shows heads. The game is won once every coin has been set aside. Writing $w_{n,p}$ for the optimal winning probability, the dynamic-programming principle gives the Bellman equation

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_{m,p}, \quad w_{0,p} := 1. \quad (1)$$

Two strategies are natural: ONE sets aside a single head each round (value $a_{n,p}$), and the greedy strategy ALL sets aside *every* head. The winning probability $b_{n,p}$ of strategy ALL is defined by the linear recursion

$$b_{n,p} = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j b_{j,p}, \quad b_{0,p} := 1, \quad (2)$$

obtained by conditioning on the number j of coins showing tails in the first round: those j coins are re-tossed, while the $n - j \geq 1$ heads are set aside for good. The companion paper [1] determines $w_{n,p}$ *exactly* at $p = \frac{1}{2}$, identifies the limit $W(p) = \lim_n w_{n,p}$ for $p > \frac{1}{2}$, and gives a first-order perturbation

expansion for $p < \frac{1}{2}$ near $\frac{1}{2}$; all of those results are machine-checked in Lean 4. The asymptotics of the greedy strategy itself are the subject of the companion paper [2]. A coin-flipping game with a related but distinct scoring rule is studied by van Doorn [3].

The present paper concerns the regime left open by a perturbative analysis: the genuine large- n behaviour of the *optimal* value for $p < \frac{1}{2}$, which turns out to be *oscillatory* rather than convergent.

1.2 Main result

Theorem 1.1 (oscillation of the optimal value). *For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone. It has infinitely many strict local maxima and infinitely many strict local minima.*

Theorem 1.1 sharpens, and corrects the natural first guess about, the first-order picture of [1]: that analysis shows $w_{n,p}$ to be eventually increasing *at first order in $\frac{1}{2} - p$* , but the exact sequence is not eventually monotone at all. The oscillation is non-perturbative — its amplitude is exponentially small in $1/(\frac{1}{2} - p)$ — which is why it is invisible to any finite-order expansion.

The proof is complete except for one step, Gap D (Section 6), flagged in the text by a boxed marker.

1.3 Strategy of proof

The proof of Theorem 1.1 has a rigid skeleton, which we record now so that the role of each later section is clear. For a real sequence (x_n) put $d_n := x_{n+1} - x_n$.

Step 0 (reduction). A bounded sequence has infinitely many strict local maxima and minima as soon as it is *not eventually monotone*, i.e. as soon as $d_n > 0$ infinitely often and $d_n < 0$ infinitely often (Lemma 4.1). Thus Theorem 1.1 splits into

(A) $(w_{n,p})$ is not eventually non-decreasing, (B) $(w_{n,p})$ is not eventually non-increasing.

Step A (the increasing side). On a stretch where w is non-decreasing, the suffix-maximum in (1) collapses to its right endpoint and the Bellman equation becomes the linear strategy-ONE recursion, whose one-step increment has the sign-definite dominant term $-q^n w_{n-1}$. Since $q > p$, this forces an eventual decrease. Statement (A) follows (Proposition 3.1); the argument is elementary and complete.

Step B (the decreasing side). On a stretch where w is non-increasing, the suffix-maximum collapses to its *left* endpoint, and w becomes — exactly — a strategy-ALL sequence (Lemma 4.2). The greedy strategy oscillates: this is the log-periodic asymptotics of the companion paper [2], recalled below as Theorem 2.1. Hence w cannot have a non-increasing tail. The transfer of the oscillation from ALL to w is carried out by a renormalisation argument (Section 5): a windowed fundamental coefficient $A(t)$ is shown to satisfy a propagation identity across the geometric scale, from which non-convergence — hence non-monotonicity — follows.

The asymmetry between Steps A and B is structural: the increasing side reduces to a contractive linear recursion with a sign-definite drift, the decreasing side to a scale-invariant recursion whose solutions oscillate. Step A is elementary; Step B carries the analytic weight of the paper.

2 Setup and notation

Throughout, $p \in (0, \frac{1}{2})$ is fixed, $q = 1 - p \in (\frac{1}{2}, 1)$, and we use freely that $q > p$, $p/q < 1$ and $2p < 1$. We write $L := \log(1/q) > 0$ and $\omega := 2\pi/L$.

The value $w_{n,p}$ satisfies (1); it is a probability, so $0 < w_{n,p} \leq 1$, and in fact $w_{n,p} < 1$ for $n \geq 1$. The two reference strategies are dominated by the optimum. Strategy ALL has winning probability $b_{n,p}$, defined by the recursion (2); strategy ONE satisfies the linear recursion

$$a_{n,p} = p^n + (1 - p^n - q^n) a_{n-1,p}, \quad a_{0,p} := 1. \quad (3)$$

In both cases $w_{n,p} \geq \max(a_{n,p}, b_{n,p})$, since the optimum dominates any fixed strategy. We use the binomial identities $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$ and $\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - q^n$. Where the value of p is fixed we abbreviate $w_n := w_{n,p}$, $b_n := b_{n,p}$.

The large- n behaviour of the greedy strategy, established in the companion paper [2], is the engine of Step B; we record it here in the form used below.

Theorem 2.1 (greedy asymptotics, [2]). *For every $p \in (0, \frac{1}{2})$ there is a continuous, strictly positive function $\Phi_p : \mathbb{R} \rightarrow \mathbb{R}$, periodic with period L , such that $b_{n,p} = \Phi_p(\log n)(1 + O(1/n))$. The first-order perturbation of $b_{n,p}$ at $p = \frac{1}{2}$ is non-constant, so that the greedy value oscillates at first order in $\frac{1}{2} - p$; it is conjectured that Φ_p is non-constant — equivalently, that $b_{n,p}$ does not converge — for every $p \in (0, \frac{1}{2})$.*

Non-constancy of Φ_p is exactly what Gap D (Section 6) requires.

3 Step A: $w_{n,p}$ is not eventually increasing

Proposition 3.1. *For $p \in (0, \frac{1}{2})$ there is no M with $w_n \geq w_{n-1}$ for all $n > M$. That is, $w_n < w_{n-1}$ for infinitely many n .*

Proof. Suppose $w_n \geq w_{n-1}$ for all $n > M$, i.e. w is non-decreasing on $\{M, M+1, \dots\}$. Fix $n > M+1$ and split the Bellman sum (1) at $j = M$, writing $S_{j,n} := \max_{j \leq m \leq n-1} w_m$. For $j \geq M$ the index window lies in the non-decreasing range, so $S_{j,n} = w_{n-1}$; for every j one has $w_{n-1} \leq S_{j,n} \leq 1$. With $U_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j$ and $T_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j S_{j,n}$,

$$w_n = p^n + T_n + (1 - p^n - q^n - U_n) w_{n-1},$$

hence

$$w_n - w_{n-1} = p^n(1 - w_{n-1}) - q^n w_{n-1} + (T_n - U_n w_{n-1}).$$

Since $w_{n-1} \leq S_{j,n} \leq 1$ one has $0 \leq T_n - U_n w_{n-1} \leq U_n$, and $U_n \leq (M-1)n^{M-1}p^{n-M+1} = C_1 n^{M-1}p^n$ with $C_1 = (M-1)p^{1-M}$. With $c := w_M > 0$ and $w_{n-1} \geq c$,

$$w_n - w_{n-1} \leq (1 + C_1 n^{M-1})p^n - c q^n.$$

As $p/q < 1$, the ratio $(1 + C_1 n^{M-1})(p/q)^n \rightarrow 0$, so the right-hand side is negative for all large n — contradicting $w_n \geq w_{n-1}$. $\square$

Proposition 3.1 is statement (A) of Section 1.3 and is fully rigorous.

4 Step B, part 1: reduction and the transfer lemma

Lemma 4.1 (reduction). *A bounded real sequence (x_n) that is neither eventually non-decreasing nor eventually non-increasing has infinitely many strict local maxima and infinitely many strict local minima, provided $x_n \neq x_{n+1}$ for all large n .*

Proof. Not eventually non-decreasing means $d_n := x_{n+1} - x_n < 0$ infinitely often; not eventually non-increasing means $d_n > 0$ infinitely often. Between a sign change of d from $-$ to $+$ lies a local minimum, between $+$ and $-$ a local maximum; infinitely many sign changes yield infinitely many of each, strict under $x_n \neq x_{n+1}$. $\square$

Lemma 4.2 (transfer to strategy ALL). *Suppose $w_n \geq w_{n+1}$ for all $n \geq M$. Define $x_0 := 1$, $x_j := \max_{j \leq m \leq M} w_m$ for $1 \leq j \leq M-1$, and $x_j := w_j$ for $j \geq M$. Then $x_n = w_n$ for $n \geq M$, and (x_n) satisfies the exact strategy-ALL recursion*

$$x_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j x_j \quad \text{for every } n > M.$$

Proof. Fix $n > M$. In the Bellman equation (1) the suffix-maximum $S_{j,n} = \max_{j \leq m \leq n-1} w_m$ collapses to the *left* endpoint on the non-increasing range: for $M \leq j \leq n-1$, $S_{j,n} = w_j = x_j$; for $1 \leq j \leq M-1$, $S_{j,n} = \max(\max_{j \leq m \leq M-1} w_m, w_M) = \max_{j \leq m \leq M} w_m = x_j$, independent of n . Writing the p^n term as $\binom{n}{0} p^n q^0 x_0$ gives the displayed recursion. $\square$

Lemma 4.2 is the crux of Step B: it says that a non-increasing tail of w is, exactly, a strategy-ALL sequence with finitely many initial values modified. The remaining task is to show such a sequence cannot exist, by transferring the oscillation of Theorem 2.1.

5 Step B, part 2: the renormalisation argument

Throughout this section (x_n) is a bounded solution of the strategy-ALL recursion for $n > M$ — either b itself, or a transferred sequence of Lemma 4.2.

5.1 The recursion as a smoothing in log-scale

With $J_n \sim \text{Bin}(n, q)$ the recursion reads $x_n = E[x_{J_n}]/(1 + q^n)$. Read on the log-scale $t := \log n$, and using $\log(qn) = \log n - L$, one application of the recursion shifts the scale down by exactly one period L and smooths by a kernel of width $\sigma(t) \sim \sqrt{p/q} e^{-t/2} \rightarrow 0$.

5.2 The local fundamental coefficient

Fix a window $\chi \in C_c^\infty(\mathbb{R})$ with $\int \chi = 1$ and $\hat{\chi}(2\pi) = 0$, and set

$$A(t) := \frac{1}{L} \sum_{n > M} x_n e^{-i\omega \log n} \chi\left(\frac{\log n - t}{L}\right) \frac{1}{n}. \quad (4)$$

This is a windowed Fourier sum at frequency ω on the multiplicative grid, the weight $1/n$ being the log-scale element $d(\log n)$. The condition $\hat{\chi}(2\pi) = 0$ makes A annihilate the constant (mean-level) component, so A extracts the fundamental oscillation. If x_n converges then $A(t) \rightarrow 0$.

5.3 The propagation estimate

Proposition 5.1 (propagation). *There are $\kappa(t) \in \mathbb{C}$ and a remainder $r(t)$ with*

$$A(t+L) = \kappa(t) A(t) + r(t), \quad |\kappa(t) - 1| = O(e^{-t}), \quad |r(t)| = O(e^{-t}).$$

The factor κ is fully under control.

Lemma 5.2 (the factor κ). *For $J_n \sim \text{Bin}(n, q)$, $\kappa(n) = E[(J_n/(qn))^i \mathbf{1}_{J_n \geq 1}] = 1 - \frac{p}{2qn}(\omega^2 + i\omega) + O(n^{-3/2})$. In particular $|\kappa - 1| = O(1/n) = O(e^{-t})$.*

Proof (sketch). Put $V_n = J_n/(qn) - 1$; the binomial central moments give $E[V_n^2] = p/(qn)$ and $E[|V_n|^3] = O(n^{-3/2})$. Expanding $(1 + V_n)^{i\omega}$ to second order on $\{|V_n| \leq \frac{1}{2}\}$ and bounding the tail by Hoeffding yields the claim. $\square$

For the remainder we pass to the *transposed* kernel. This makes the negative-binomial structure of the recursion explicit and, as it turns out, removes the need for any limit theorem. Write $a_n(t) := n^{-1} e^{-i\omega \log n} \chi\left(\frac{\log n - t}{L}\right)$, so that $A(t) = \frac{1}{L} \sum_{n > M} x_n a_n(t)$.

Lemma 5.3 (transposed kernel and the negative binomial). *Substituting the recursion $x_n = (1 + q^n)^{-1} \sum_{j=0}^n P(\text{Bin}(n, q) = j) x_j$ into $A(t+L)$ and exchanging the (finite) order of summation,*

$$A(t+L) = \frac{1}{L} \sum_j x_j W(j, t), \quad W(j, t) = \sum_{n \geq j} \frac{P(\text{Bin}(n, q) = j)}{1 + q^n} a_n(t+L).$$

The transposed weight has exact total mass

$$\sum_{n \geq j} P(\text{Bin}(n, q) = j) = q^j \sum_{k \geq 0} \binom{j+k}{j} p^k = q^j (1-p)^{-(j+1)} = \frac{1}{q},$$

and $q P(\text{Bin}(n, q) = j)$ is the law of $N_j := j + \text{NB}(j+1, q)$, with

$$E[N_j] = \bar{n}_j := \frac{j+p}{q}, \quad \text{Var}(N_j) = \frac{(j+1)p}{q^2}.$$

Consequently $W(j, t) = \frac{1}{q} E[(1 + q^{N_j})^{-1} a_{N_j}(t+L)]$.

Proof. The substitution is valid because the window lies far above M ; the swap is exact since χ has compact support, so every sum is finite. The indices $j \leq M$ contribute super-exponentially little ($P(\text{Bin}(n, q) = j)$ is super-exponentially small for fixed j and n in the window) and are absorbed silently. The mass identity is the negative-binomial series $\sum_{k \geq 0} \binom{j+k}{k} p^k = (1-p)^{-(j+1)}$; hence $q^{j+1} \binom{j+k}{k} p^k$, $k \geq 0$, is the law of $\text{NB}(j+1, q)$ in $k = n - j$, and its mean and variance, shifted by j , are those stated. $\square$

The exact identity is the crux. The transposed weight carries mass *exactly* $1/q$ and is a genuine probability law — being a shifted sum of i.i.d. geometrics, an exceptionally well-behaved one. The oscillatory cancellation is delivered by the recursion's own combinatorial structure; no local limit theorem enters.

Proof of Proposition 5.1. The transposed picture settles κ and r together: we show $A(t+L) = A(t) + O(e^{-t})$, and then $r(t) = A(t+L) - \kappa(t)A(t) = (1 - \kappa(t))A(t) + O(e^{-t}) = O(e^{-t})$, using $|\kappa - 1| = O(e^{-t})$ (Lemma 5.2) and the boundedness of A .

For real $n > 0$ the map $n \mapsto a_n(t+L)$ is smooth, supported in the geometric window $\log n \in t+L+L \operatorname{supp} \chi$, and every derivative obeys $|a_n^{(k)}(t+L)| \leq C_k n^{-k-1}$: each n -derivative produces one factor n^{-1} and bounded constants $(\omega, L^{-1}, \|\chi^{(k)}\|_\infty)$. The factor $(1+q^n)^{-1} = 1 + O(q^n)$ perturbs this super-exponentially little.

Fix j with $\bar{n}_j$ within $O(1)$ of the window, so j is of order e^t . A second-order Taylor expansion of $a_n(t+L)$ about $n = \bar{n}_j$, taken under $E[\cdot]$, loses its first-order term because $E[N_j - \bar{n}_j] = 0$:

$$E[a_{N_j}(t+L)] = a_{\bar{n}_j}(t+L) + \frac{1}{2} E[a_{\xi_j}''(t+L) (N_j - \bar{n}_j)^2],$$

with ξ_j between N_j and $\bar{n}_j$, hence $\xi_j \geq j$ (both $N_j \geq j$ and $\bar{n}_j > j$). Thus $|a_{\xi_j}''| \leq C_2 j^{-3}$ and, since $\operatorname{Var}(N_j) = O(j)$,

$$|E[a_{N_j}(t+L)] - a_{\bar{n}_j}(t+L)| \leq \frac{1}{2} C_2 j^{-3} \operatorname{Var}(N_j) = O(j^{-2}).$$

With the super-exponentially small $(1+q^{N_j})^{-1}$ correction, $W(j, t) = \frac{1}{q} a_{\bar{n}_j}(t+L) + O(j^{-2})$.

It remains to undo the period shift. From $\bar{n}_j = (j+p)/q$,

$$\log \bar{n}_j = \log j + L + \log(1+p/j), \quad e^{-i\omega L} = 1,$$

and $\log(1+p/j) = O(j^{-1})$; expanding the three factors of $a_{\bar{n}_j}(t+L)$ gives $\frac{1}{q} a_{\bar{n}_j}(t+L) = a_j(t) (1 + O(j^{-1})) = a_j(t) + O(j^{-2})$, the last step because $a_j(t) = O(j^{-1})$. Hence $W(j, t) = a_j(t) + O(j^{-2})$ uniformly in j , and

$$A(t+L) - A(t) = \frac{1}{L} \sum_j x_j [W(j, t) - a_j(t)], \quad |A(t+L) - A(t)| \leq \frac{\|x\|_\infty}{L} \sum_j O(j^{-2}).$$

The window confines j to $[e^{t-cL}, e^{t+cL}]$ with c fixed by $\operatorname{supp} \chi$; all these indices exceed e^{t-cL} , so $\sum_j O(j^{-2}) = O(e^{-t})$. This proves $A(t+L) = A(t) + O(e^{-t})$, and with it Proposition 5.1. $\square$

In an earlier form the remainder bound was a separate gap — a local central limit theorem, with one Edgeworth correction, for the binomial log-deviation density. The transposed picture closes it: the cancellation is supplied by the *exact* negative-binomial identity of Lemma 5.3, and the windowed coefficient obeys the elementary propagation law $A(t+L) = A(t) + O(e^{-t})$.

5.4 From propagation to non-convergence

Theorem 5.4 (renormalisation). *Let (x_n) be a bounded solution of the strategy-ALL recursion for $n > M$. Then for each t_0 the sequence $A(t_0 + kL)$ converges, as $k \rightarrow \infty$, to a limit A_∞ . Moreover: if x_n converges then $A_\infty = 0$; and if $A_\infty \neq 0$ then (x_n) is not eventually monotone.*

Proof. Write $A_k = A(t_0 + kL)$, $\kappa_k = \kappa(t_0 + kL)$, $r_k = r(t_0 + kL)$. By Proposition 5.1, $\sum_k |\kappa_k - 1| < \infty$ and $\sum_k |r_k| < \infty$ (geometric series in $e^{-L} < 1$). Solving $A_{k+1} = \kappa_k A_k + r_k$,

$$A_k = \left(\prod_{j < k} \kappa_j \right) A_0 + \sum_{j < k} \left(\prod_{j < i < k} \kappa_i \right) r_j \longrightarrow A_\infty,$$

the products and series converging absolutely. If x_n converges then $A(t) \rightarrow 0$, so $A_\infty = 0$. If $A_\infty \neq 0$ then $A_k \not\rightarrow 0$, so x_n does not converge; a bounded non-convergent sequence is not eventually monotone. $\square$

6 The base case and the proof of Theorem 1.1

Theorem 5.4 reduces Step B to a single non-degeneracy: the limit A_∞ must be non-zero.

Proposition 6.1 (base case). *For every transferred sequence (x_n) of Lemma 4.2, $A_\infty \neq 0$.*

By linearity $A_\infty(x) = A_\infty(b) + A_\infty(g)$, where $g := x - b$ satisfies $g \geq 0$ (the optimum dominates strategy ALL) and solves the strategy-ALL recursion for $n > M$.

► **OPEN GAP — Gap D — the non-degeneracy. Work continues here.**

$A_\infty(x) \neq 0$ is the genuine arithmetic heart of the proof. It needs two things: that pure greedy play oscillates — $A_\infty(b) \neq 0$, the non-constancy of Φ_p — and that the perturbation g does not cancel that oscillation, $A_\infty(g) \neq -A_\infty(b)$. The first holds at first order near $p = \frac{1}{2}$ and is conjectured throughout $(0, \frac{1}{2})$ (Theorem 2.1, from [2]); the second is not secured by $g \geq 0$ alone, and must draw on the optimality of w . Work continues here.

Proof of Theorem 1.1. Fix $p \in (0, \frac{1}{2})$. By Lemma 4.1 it suffices to show that $(w_{n,p})$ is neither eventually non-decreasing nor eventually non-increasing.

Not eventually non-decreasing is Proposition 3.1.

Not eventually non-increasing. Suppose, for contradiction, $w_n \geq w_{n+1}$ for all $n \geq M$. By Lemma 4.2 the transferred sequence (x_n) satisfies $x_n = w_n$ for $n \geq M$ and solves the strategy-ALL recursion for $n > M$; being a tail of a monotone bounded sequence, x_n converges. By Theorem 5.4 then $A_\infty = 0$, contradicting Proposition 6.1. Hence no non-increasing tail exists.

Both one-sided statements hold, so $(w_{n,p})$ is not eventually monotone; Lemma 4.1 gives infinitely many strict local maxima and minima. $\square$

Remark 6.2 (strictness). Lemma 4.1 delivers *strict* extrema once $w_{n,p} \neq w_{n+1,p}$ for all large n , which holds in all computed data and is expected for every $p \in (0, \frac{1}{2})$; absent it, Theorem 1.1 holds with the non-strict notion of local extremum.

7 Numerical evidence

The renormalisation argument has been checked numerically in arbitrary-precision arithmetic (**simulation/renorm**). the windowed coefficient $A(t_0 + kL)$ is observed to converge to a non-zero limit A_∞ for every tested $p \in \{0.35, 0.40, 0.42, 0.45\}$, and $n|\kappa(n) - 1|$ converges to the constant $\frac{p}{2q}|\omega^2 + i\omega|$ of Lemma 5.2 — a direct quantitative confirmation of Proposition 5.1. The log-periodic asymptotics of the greedy strategy that feed the argument are verified separately in [2].

8 Status and open problems

The proof skeleton is complete and the rigorous backbone is substantial. The reduction (Lemma 4.1), the elementary Step A (Proposition 3.1), the exact transfer lemma (Lemma 4.2), the factor estimate (Lemma 5.2), the propagation estimate (Proposition 5.1, via the transposed negative-binomial kernel and the exact identity $\sum_{n \geq j} P(\text{Bin}(n, q) = j) = 1/q$), and the renormalisation theorem (Theorem 5.4) are all rigorous; the log-periodic asymptotics of the greedy strategy are supplied by the companion paper [2]. A single gap remains open; it is flagged in the text and summarised here.

Gap	What is missing	Where
D	the non-degeneracy $A_\infty(x) \neq 0$ (transferred sequence)	§6

A complete proof of Theorem 1.1 consists of Proposition 3.1 (done) together with Gap D alone. Gap D, the non-degeneracy $A_\infty(x) \neq 0$, has two parts. (i) The greedy strategy oscillates: $A_\infty(b) \neq 0$, the non-constancy of Φ_p . By [2] this is settled at first order near $p = \frac{1}{2}$ and conjectured throughout $(0, \frac{1}{2})$. (ii) The optimal perturbation $g = w - b \geq 0$ does not cancel that oscillation. Positivity of g alone does not secure (ii): one checks that the limits A_∞ of non-negative strategy-ALL solutions wind around the origin, so some non-negative g *does* cancel. Excluding this for the *optimal* sequence must use the optimality of w — it is a supersolution of the strategy-ALL recursion — not merely $w \geq b$. That is the current front of work.

References

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- [3] E. A. van Doorn, *Optimal play of a coin-flipping game*, preprint, arXiv:2406.14700, 2024.